MEDINOVA TEACHING HOSPITAL

Class 13 — Surgery & Perioperative Care

Preoperative Assessment · Surgical Emergencies · Wound Management · Anastomotic Complications

> *"Surgery begins and ends with the physician, not the surgeon."*

FELLOWSHIP SERIES | MODULE 13 of 20

SECTION 01

Preoperative Risk Stratification

ASA Physical Status Classification

ASA	Description	Clinical Example
I	Healthy patient	Non-smoker, minimal alcohol use.
II	Mild systemic disease	Well-controlled HTN, DM, or Smoker.
III	Severe systemic disease	Poorly controlled DM/HTN, COPD, BMI > 40.
IV	Constant threat to life	Recent MI (< 3 months), CVA, TIA, or Sepsis.
V	Moribund; not expected to survive	Ruptured AAA, massive trauma, intracranial bleed.

Scoring Tools

- **POSSUM / P-POSSUM:** Physiological and Operative Severity Score for the enumeration of Mortality and Morbidity. Used for audit and predicting individual risk.
- **CPET:** Cardiopulmonary Exercise Testing (VO2 max) — the gold standard for objective functional capacity assessment.

> **Prof. Obi's Pearl #1** "The ASA score is a snapshot of physiology, not a diagnosis. A high ASA score tells you to spend more time in the anesthetic room, not necessarily to cancel the case."

SECTION 02

Acute Surgical Emergencies

Alvarado Score (MANTRELS) for Appendicitis

Criteria	Points
Migration of pain | **A**norexia | **N**ausea	1 each
Tenderness in RIF	2 points
Rebound tenderness | **E**levated Temp (>37.3°C)	1 each
Leukocytosis (>10 x 10^9/L)	2 points
Shift to the left (Neutrophilia)	1 point

Score 5-6: Possible | 7-8: Probable | 9-10: Definite Appendicitis.

Bowel Obstruction & Ischaemia

- **Small Bowel (SBO):** Vomiting (early), Colicky pain, Absolute constipation (late).
- **Mesenteric Ischaemia:** "Pain out of proportion to findings." High lactate is a late and ominous sign.

> **Prof. Obi's Pearl #2** "Never let the sun set or rise on a complete bowel obstruction. But remember: 'Drip and Suck' (IV fluids and NGT) buys you time to decide if the bowel is dying or just sleeping."

SECTION 03

Perioperative Medication Management

Drug Class	Action on Day of Surgery	Rationale
ACEi / ARBs	**HOLD**	Risk of profound intraoperative hypotension.
Beta-Blockers	CONTINUE	Prevents rebound tachycardia/ischaemia.
Warfarin	**HOLD (5 Days)**	Target INR < 1.5. Consider bridging with LMWH.
DOACs	**HOLD (24-48h)**	Based on renal function and surgical risk.
SGLT2i	**HOLD (2 Days)**	Risk of euglycaemic DKA.
Steroids	CONTINUE	Requires stress-dose hydrocortisone IV.

SECTION 04

Postoperative Complications & Timing

- **Day 1-2 (Wind):** Atelectasis, Pneumonia.
- **Day 3-5 (Water):** Urinary Tract Infection (UTI).
- **Day 5-7 (Walk):** DVT / PE.
- **Day 7-10 (Wound):** Surgical Site Infection (SSI), Anastomotic Leak.
- **Anytime (Wonder):** Drugs / Transfusion reactions.

SECTION 05

Wound Management Classification

Class	Type	Example
I	Clean	No entry into viscera (e.g., Hernia repair).
II	Clean-Contam	Controlled entry into viscera (e.g., Cholecystectomy).
III	Contaminated	Gross spillage/inflammation (e.g., Perforated ulcer).
IV	Dirty	Pus/Faecal contamination (e.g., Abscess drainage).

SECTION 06

Surgical Nutrition

Refeeding Syndrome: Low Phosphate, Low Magnesium, Low Potassium. Occurs when feeding a malnourished patient too quickly.

- **Enteral (EN):** Always preferred if the gut works ("Use it or lose it").
- **Parenteral (TPN):** Only if the gut is non-functional (e.g., high-output fistula, short bowel).

> Prof. Obi's Pearl #3 "A patient with a serum albumin < 30 g/L is a surgical ticking time bomb. They won't heal their wound, and they certainly won't heal their anastomosis. Feed them before you cut them."

SECTION 07

Enhanced Recovery After Surgery (ERAS)

Key Principles:
- **Pre-op:** Carbohydrate loading (no prolonged fasting), Counseling.
- **Intra-op:** Mid-thoracic epidural, Goal-directed fluid therapy, Minimally invasive.
- **Post-op:** Early mobilization (out of bed Day 0), Early oral intake, Avoidance of opiates.

> **Prof. Obi's Pearl #4** "The best way to manage a surgical complication is to prevent it. But the second best way is to acknowledge it early. A 'tachycardia' on Day 5 is an anastomotic leak until you prove it's just pain."

Medical Definitions:
- **Anastomosis:** The surgical connection between two structures (e.g., bowel loops).
- **Ileus:** Temporary paralysis of bowel motility post-surgery.
- **Atelectasis:** Collapse of lung alveoli due to shallow breathing/pain.
- **Primary Intention:** Closing a wound with sutures/staples immediately.

OFFICIAL SIGN-OFF

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